Guan-Yu Chen

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Profile

AI application engineer and data science M.S. student focused on LLM-backed services, RAG/knowledge systems, AI agents, evaluation-ready workflows, and privacy-preserving AI. Hands-on experience spans TSMC DTP semiconductor design automation, NTU medical AI research, federated learning with NVIDIA FLARE, and GPU/HPC performance engineering.

Education

National Taiwan University (With Academia Sinica) *Sep. 2024 – Present*
M.S. in Data Science (College of Electrical Engineering & Computer Science)
- **GPA:** 4.22/4.30 (**Rank:** 1/13, 46 Credits)
- **Laboratory:** AI Application and Integration Lab (Dept. CSIE, NTU)

Chang Gung University *Sep. 2020 – Jun. 2024*
B.S. in Computer Science and Information Engineering (College of Engineering)
- **GPA:** 3.94/4.00 (**Rank:** 1/67, 160 Credits, Valedictorian)
- **Laboratory:** AI and Multimedia Applications Lab (Dept. CSIE, CGU)

Experience

R&D Intern, TSMC, Design Technology Platform (DTP) *Jul. 2025 – Aug. 2025*
Python (PyTorch, Unsloth, LangChain) | LoRA | AI Agents/RAG | EDA Automation
- Top 10 Finalist, 2025 DTP Intern Conference for project: "LLM-Driven Automation Framework for LVS Deck Coding Copilot and Layout Pattern Constraint Generation".
- Built public-safe AI workflow components for EDA automation scenarios, combining multi-agent orchestration, LoRA-tuned LLM behavior, and structured specification/config generation.
- Focused on production-readiness concerns for proprietary engineering data: traceable outputs, human review loops, reusable prompts/procedures, and clear data-boundary assumptions.

Research Assistant, National Taiwan University, Department of CSIE *Sep. 2024 – Present*
Python (PyTorch, CUDA, cuDNN, OpenCV) | NVIDIA FLARE | Docker | CVAT
- Researching continual test-time adaptation and parameter-efficient adaptation for robust AI under distribution shift and long-horizon deployment.
- Developed segmentation models for NTUH's handheld ultrasound pipeline, connecting model quality with real-time medical imaging constraints.
- Integrated NVIDIA FLARE-based federated learning collaboration with NCKUH, supporting privacy-preserving distributed training without centralizing sensitive data.

Research Assistant, Chang Gung Memorial Hospital *Jan. 2024 – Jan. 2025*
Java (Android Studio, OpenCV) | Python (TensorFlow, Keras, Scikit-learn) | Firebase
- Developed an Android/computer-vision workflow for fine motor skill assessment and digital case management, improving process efficiency by over 50%.
- Co-designed a standardized BOT-2-based digital assessment workflow with physicians, improving reliability, consistency, and fine motor skill evaluation accuracy by 20%.
- Optimized stroke fitting using quadratic Bézier curves, achieving $\mathcal{O}(n)$ time complexity for real-time processing.

Projects

Smart Healthcare Dialogue System for ERAS Support *2025*
Python (Flask, SQLite) | LLM (Qwen) | RAG | LoRA | Evaluation
- Built an on-premises healthcare chatbot concept for ERAS, using multilingual RAG and LoRA to support clinical Q&A and personalized care workflows.
- Designed a modular pipeline with semantic knowledge base, QA/VQA benchmarks, and automated plus expert evaluation for clinical reliability.

OVSS Continual Test-Time Adaptation Research Platform *2026*
Python (PyTorch) | CLIP/NACLIP | DINOv2 | Segmentation | Experiment Pipelines
- Developed long-horizon open-vocabulary semantic segmentation experiments for continual domain shift, targeting stability without source data or labels.

- Implemented CLIP/NACLIP adaptation, DINOv2-based evaluation refinement, and ACDC/OnDA benchmark pipelines for robust AI evaluation.

High-Performance Parallel Computing & GPU Acceleration *2025*

C/C++ | CUDA | MPI | OpenMP | AVX-512 | Nsight Compute

- Optimized CUDA kernels with shared memory and padding to reduce shared-memory bank conflicts, analyzing performance metrics via Nsight Compute.
- Implemented distributed attention and convolution algorithms using MPI/OpenMP, improving throughput through AVX-512 vectorization and memory alignment.